



Modelling Financial Markets



Seminar 3: Factor Models and Asset Pricing — Hands-on

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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FM: Fama–MacBeth (regressions)
- ▣ FWER: Family-Wise Error Rate
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ MOM: MOMentum factor
- ▣ NW: Newey–West (standard errors)
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ SMB: Small Minus Big (size factor)
- ▣ SML: Security Market Line
- ▣ US: United States

Outline

Introduction

Key Definitions

Part A: Pen-and-Paper Exercises

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the CAPM from the tangency portfolio and compute portfolio betas
- ▣ Estimate betas and alphas with robust and bootstrap inference
- ▣ Test the CAPM jointly (GRS) and estimate risk premia (Fama–MacBeth)
- ▣ Correct for multiple testing and extract latent factors with PCA

- ▣ Background: Lecture 3; textbook: Campbell, Lo & MacKinlay (1997), ch. 5–6
- ▣ Seminar: Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises, US and Romanian data (B1–B8)
- ▣ **Part C:** one open question, a starting point for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- Report every estimate with its uncertainty (standard error, interval or p -value)
- Before any regression: align the calendars and look for data errors (unadjusted corporate events)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (CAPM from the tangency portfolio), A3 (Vasicek beta), A5 (Bonferroni, Holm, BH, step by step), A7 (PCA of two assets)
- Part B: B1 (beta and alpha of XLK with HAC and bootstrap), B2 (GRS on sector ETFs), B3 (PCA of sector ETFs)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A3); A6 (model: A5); A8 (model: A7)
- Part B: B4, B6 (model: B1); B5 (model: B2 and Lecture 3, Section 5); B7 (model: A5); B8 (model: B3)
- Part C: C1, open question (model: B1, B5); C2, critique an AI answer (model: C1)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter3_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`

```
import numpy as np, pandas as pd
URL = "https://raw.githubusercontent.com/danpele/MFM/main/data/market/"
x1k = pd.read_csv(URL + "XLK.US.csv", index_col="date",
                  parse_dates=True)["adjusted_close"]
FF = "https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/"
# F-F_Research_Data_5_Factors_2x3_CSV.zip
```

- ETFs and stocks: `adjusted_close`; BET: `close`; factors: [Kenneth French Data Library](#) (percent, divide by 100)
- Joint analyses: join the *prices* on common days first, then take returns

Key Definitions: CAPM and Beta

- ▣ **Excess return** $R_{i,t}^e = R_{i,t} - R_{f,t}$; **CAPM** (Sharpe (1964)): $\mathbb{E}[R_i^e] = \beta_i \mathbb{E}[R_m^e]$
- ▣ **Beta** $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$; **Jensen's alpha**: intercept of $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$
- ▣ **Portfolio beta** $\beta_p = \sum_i w_i \beta_i$ (betas are linear in the weights)
- ▣ **Blume** (Blume (1971)): $\beta^B = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek** (Vasicek (1973)): $\beta^V = w \hat{\beta} + (1 - w) \bar{\beta}$, $w = \sigma_\beta^2 / (\sigma_\beta^2 + \text{se}(\hat{\beta})^2)$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) standard errors, Newey & West (1987): Bartlett weights $1 - l / (L + 1)$

Key Definitions: Tests and Latent Factors

- ▣ **GRS** (Gibbons, Ross & Shanken (1989)): $W = \frac{T-N-1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$ under H_0 : all $\alpha_i = 0$
- ▣ **Fama–MacBeth** (Fama & MacBeth (1973)): monthly cross-sectional slopes $\hat{\lambda}_t$, averaged; **Shanken** (Shanken (1992)) inflates the variance by $(1 + \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda})$
- ▣ **Bonferroni**: $p_j \leq \alpha/M$; **Holm** (Holm (1979)): $p_{(j)} \leq \alpha/(M - j + 1)$ step by step; **BH** (Benjamini & Hochberg (1995)): largest j with $p_{(j)} \leq j\alpha/M$
- ▣ **PCA** (Principal Component Analysis): eigenvalues λ_k and eigenvectors of the correlation matrix; share λ_k/N
- ▣ **Moving-block bootstrap** (Künsch (1989)): resample blocks of consecutive observations, keeping the dependence inside each block

Exercise A1: The CAPM from the Tangency Portfolio [Solved]

Task

- ▣ The tangency portfolio is $w_m = \Sigma^{-1}\mu / (1^\top \Sigma^{-1}\mu)$, μ the expected excess returns
- ▣ Assume it is the market portfolio: show that $\mu_i = \beta_i \mu_m$ for every asset
- ▣ Hint: compute $\text{Cov}(R, R_m)$ as a vector

Solution

- ▣ Let $c = 1^\top \Sigma^{-1}\mu$; then $\Sigma w_m = \mu / c$
- ▣ $\text{Cov}(R, R_m) = \Sigma w_m = \mu / c$
- ▣ $\sigma_m^2 = w_m^\top \Sigma w_m = w_m^\top \mu / c = \mu_m / c$
- ▣ Hence $\mu = c \text{Cov}(R, R_m) = \mu_m \text{Cov}(R, R_m) / \sigma_m^2 = \beta \mu_m$
- ▣ The SML is a property of **any** efficient portfolio (Roll's critique)

Exercise A2: Portfolio Beta [Proposed]

Solution discussed in class

Task

- Monthly betas 1999–2026 (Exercise B2): XLK 1.277, XLU 0.401, XLF 1.068
- Portfolio: 50% XLK, 30% XLU, 20% XLF. Compute β_p and the CAPM expected excess return with a premium of 7.19%
- Compute the beta of long XLK, short XLU (100% each). Is it market-neutral?
- Model: A1

Exercise A3: Vasicek Beta of XLK [Solved]

Task

- XLK, monthly 1999–2012: $\hat{\beta} = 1.415$, $\text{se}(\hat{\beta}) = 0.139$
- Cross-section of the 9 sector betas: mean $\bar{\beta} = 0.949$, variance $\sigma_{\bar{\beta}}^2 = 0.116$
- Compute the Vasicek and the Blume betas; compare with the realised beta in 2013–2026, 1.172

Solution

- $w = 0.116 / (0.116 + 0.0193) = 0.857$
- $\beta^V = 0.857(1.415) + 0.143(0.949) = 1.348$
- $\beta^B = 0.33 + 0.67(1.415) = 1.278$
- Errors vs 1.172: raw 0.243, Vasicek 0.176, Blume 0.106
- Vasicek shrinks little because XLK's beta is precise; Blume shrinks everything by a third

Exercise A4: Blume Forecasts for Defensive Sectors [Proposed]

Solution discussed in class

Task

- ▣ Regression of 2013–2026 betas on 1999–2012 betas of 9 sectors: $\hat{\beta}_2 = 0.230 + 0.724 \hat{\beta}_1$
- ▣ Forecast the later betas of XLP ($\hat{\beta}_1 = 0.451$) and XLU ($\hat{\beta}_1 = 0.463$), with the fitted line and with Blume's $0.33 + 0.67\hat{\beta}$
- ▣ Realised: XLP 0.578, XLU 0.455. Which forecast is better, and why can shrinkage fail for one sector?
- ▣ Model: A3

Exercise A5: Ten Candidate Factors [Solved]

Task

- ▣ Ten new factors have p -values
 - ▶ 0.001, 0.004, 0.009, 0.012, 0.021
 - ▶ 0.035, 0.048, 0.060, 0.20, 0.55
- ▣ At $\alpha = 5\%$, how many are discoveries: naive, Bonferroni, Holm, Benjamini–Hochberg?
- ▣ Which error rate does each method control?

Solution

- ▣ Naive ($p \leq 0.05$): 7
- ▣ Bonferroni ($p \leq 0.005$): 2
- ▣ Holm: thresholds 0.005, 0.0056, 0.00625; $p_{(3)} = 0.009$ fails: 2
- ▣ BH: thresholds $0.005j$; $p_{(5)} = 0.021 \leq 0.025$, $p_{(6)} = 0.035 > 0.030$ and no later success: 5
- ▣ Bonferroni and Holm control FWER; BH controls FDR

Exercise A6: 300 Useless Factors [Proposed]

Solution discussed in class

Task

- ▣ 300 independent factors, none with a true premium; each t -statistic follows the standard Normal distribution
- ▣ Expected number with $|t| > 1.96$? With $|t| > 3$?
- ▣ Probability that at least one has $|t| > 3$?
- ▣ Bonferroni threshold for $|t|$ at a 5% FWER
- ▣ Model: A5

Exercise A7: PCA of Two Assets [Solved]

Task

- ▣ Two standardised returns with correlation $\rho = 0.6$
- ▣ Find the eigenvalues and eigenvectors of the correlation matrix
- ▣ What share of variance does PC1 explain? What happens when $\rho \rightarrow 1$?

Solution

- ▣ $\det \begin{pmatrix} 1 - \lambda & \rho \\ \rho & 1 - \lambda \end{pmatrix} = 0 \Rightarrow \lambda = 1 \pm \rho$
- ▣ $\lambda_1 = 1.6$ with $(1, 1)/\sqrt{2}$; $\lambda_2 = 0.4$ with $(1, -1)/\sqrt{2}$
- ▣ PC1 share = $1.6/2 = 80\%$
- ▣ $\rho \rightarrow 1$: one factor explains everything; $\rho = 0$: no common factor

Exercise A8: Equicorrelated Assets [Proposed]

Solution discussed in class

Task

- ▣ Three standardised returns, all pairwise correlations $\rho = 0.5$
- ▣ Find the eigenvalues and the PC1 share
- ▣ Generalise to N assets: what share does PC1 reach as $N \rightarrow \infty$?
- ▣ Model: A7

Exercise B1: Beta and Alpha of XLK [Solved]

Task

- Monthly excess returns of XLK (month-end prices) on Mkt-RF, Jan 1999 – Jul 2026
- Report $\hat{\alpha}$ (annualised) and $\hat{\beta}$ with classical and Newey–West (6 lags) errors
- 95% moving-block bootstrap interval for α (blocks of 12 months, 2000 draws)

Interpretation

- The HAC error of beta is twice the classical one: which feature of the residuals causes it?

Results ($T = 331$)

	Est.	Classical	NW
α (ann.)	0.33%	2.27%	2.50%
β	1.277	0.041	0.085

- $R^2 = 0.742$; $t(\alpha) = 0.13$
- Bootstrap: $\alpha \in [-5.4\%, 4.8\%]$, s.e. 2.61%
- Residual variance moves with market volatility (tech bust, 2008, 2020): heteroskedasticity inflates the error of beta

Exercise B2: GRS Test on Sector ETFs [Solved]

Task

- 9 sector ETFs, monthly excess returns 1999–2026, market Mkt-RF
- Estimate the 9 alphas and betas; compute GRS and its p -value

Interpretation

- The same test rejects the CAPM on the 25 size $\times$ B/M portfolios (GRS = 4.18). Why is it silent here?

Results ($T = 331$, $N = 9$)

- GRS = 0.53, $p = 0.85$: do not reject
- Alphas: XLU +3.4%, XLE +2.4%, XLV +1.8%, XLP +1.7%, XLF -2.3%
- Betas: XLU 0.40, XLP 0.45, XLV 0.67, XLK 1.28
- Sectors do not sort on size or value: little dispersion in the alphas that matter, hence low power
- Not rejecting $\neq$ accepting the CAPM

Exercise B3: PCA of Sector ETFs [Solved]

Task

- 11 sector ETFs, daily log returns on common days, June 2018 – September 2026
- Standardise, compute the correlation matrix, its eigenvalues and loadings
- Correlate PC1 with SPY

Interpretation

- PC2 opposes utilities and staples to technology: would it be priced (earn a premium)? What would you need to test it?

Results ($T = 2073$)

- Shares: PC1 66.0%, PC2 9.7%, first three 83.1%
- Two eigenvalues above 1
- $\text{corr}(\text{PC1}, \text{SPY}) = 0.954$
- PCA explains covariance, not average returns: a premium needs a cross-sectional test (Fama–MacBeth on PC2 loadings) or a long–short portfolio and its mean

Exercise B4: CAPM Betas of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- ▣ 11 blue chips vs BET, daily log returns from Jan 2015 or listing; R_f assumed constant
- ▣ Find and remove data errors ($|r_t| > 0.5$); report $\hat{\beta}$, Newey–West errors, R^2 and Vasicek betas
- ▣ Model: B1, A3

Interpretation

- ▣ How would asynchronous trading of less liquid stocks bias daily betas? How would you check it?

 MFM_ch3_bvb_betas

Exercise B5: Fama–MacBeth on ETFs [Proposed]

Solution discussed in class

Task

- ▣ Test assets: 9 sector and 7 factor ETFs, monthly excess returns Aug 2013 – Jul 2026
- ▣ Factors: Mkt-RF, SMB, HML. Run both passes; report $\hat{\lambda}$ with Fama–MacBeth, Newey–West and Shanken errors
- ▣ Compare $\hat{\lambda}$ with the time-series factor means
- ▣ Model: B2 and Lecture 3, Section 5

Interpretation

- ▣ Is a non-significant premium here evidence that size and value are not priced?

Exercise B6: Do Factor ETFs Have Alpha? [Proposed]

Solution discussed in class

Task

- Daily excess returns of MTUM and QUAL on FF5 + MOM (Fama & French (2015), Carhart (1997)), each on its own calendar, only between consecutive factor trading days
- Annualised alpha with Newey–West (10 lags) and a moving-block bootstrap interval (blocks of 20 days, 1000 draws)
- Model: B1

Interpretation

- Why must a 3-day ETF return never be regressed on a 1-day factor return?

Exercise B7: Multiple Testing of 25 Alphas [Proposed]

Solution discussed in class

Task

- ▣ FF3 (Fama & French (1993)) time-series regressions of the 25 size $\times$ B/M portfolios, monthly 1963–2026
- ▣ Newey–West t -statistics of the 25 alphas and two-sided p -values
- ▣ Count discoveries: naive, Bonferroni, Holm, BH; and $|t| > 3$ (Harvey, Liu & Zhu (2016))
- ▣ Model: A5

Interpretation

- ▣ The 25 tests are strongly correlated: which methods remain valid?

 MFM_ch3_multiple_testing

Exercise B8: PCA of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- ▣ 10 blue chips (H2O excluded: listed in 2023), daily log returns on common days from June 2017
- ▣ Remove days with a data error; compute eigenvalue shares and PC1 loadings; correlate PC1 with BET
- ▣ Compare the PC1 share with the US sectors (B3)
- ▣ Model: B3

Interpretation

- ▣ What does a weaker first factor imply for diversification within the BVB?

 MFM_ch3_bvb_pca

C1: Do Factor Premia Exist on the Bucharest Stock Exchange? [Proposed]

Solution discussed in class

The question

- Do momentum (Jegadeesh & Titman (1993)) and low beta (Frazzini & Pedersen (2014)) earn premia among BVB blue chips?
- Build monthly long–short portfolios from the blue chips and test their means
- Size and value need fundamentals (book equity, shares outstanding): which public sources could you use?

Deliverable

- A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- Signals known at the start of each month (no look-ahead)
- Mean, volatility and a Newey–West t of each long–short return
- A discussion of the number of stocks and of short-selling constraints
- The $|t| > 3$ hurdle of the factor zoo
- Transaction costs and liquidity

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “P holds cleaned month-end adjusted prices of the BVB blue chips of C1, one column per stock. Build a 12-month momentum long–short portfolio, top 3 minus bottom 3 with equal weights, and report its annual mean and volatility”
- **The answer:** the code below, followed by the conclusion

```
R      = np.log(P).diff()                # monthly returns
mom    = P.pct_change(12)                # past 12-month return
ls     = {}
for t in R.index[13:]:
    s = mom.loc[t].dropna().sort_values() # rank the stocks
    ls[t] = R.loc[t, s.index[-3:]].mean() - R.loc[t, s.index[:3]].mean()
ls = pd.Series(ls)
print(12 * ls.mean(), np.sqrt(12) * ls.std()) # annual mean, volatility
```

- **Conclusion of the AI:** “Momentum earns about 49% a year with a volatility of 16%. This agrees with [Fama & French \(2012\)](#), who document momentum on the Bucharest Stock Exchange.”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- (a) Find the errors: there are two or three; the rest of the answer is correct
- (b) For each error, say how you would detect it: a test, a published number, the lecture or the paper itself
- (c) Give the corrected code and the numbers it produces
- (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Model: C1 and Lecture 3

Why this exercise

- AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

Data

- ▣ Unadjusted corporate events (spin-offs, capital returns) create fake returns
- ▣ Regressing multi-day asset returns on one-day factors
- ▣ Forgetting that French data are in percent
- ▣ Using signals that were not known at the portfolio date

Statistics

- ▣ Classical errors for betas under volatility clustering
- ▣ Fama–MacBeth errors without the Shanken correction
- ▣ $|t| > 1.96$ after trying many factors
- ▣ Reading “not rejected” as “true”

- ▣ Rule: state the sample, the frequency, the market proxy and the risk-free rate of every regression

Take-Home Assignment (Team Project)

- ▣ Choose **five** assets: at least two BVB stocks, one US sector ETF and one factor ETF
- ▣ For each asset
 - ▶ CAPM beta with Newey–West errors, a rolling beta and its Vasicek-adjusted value
 - ▶ For US assets: FF5 + MOM loadings and alpha with a bootstrap interval
- ▣ Jointly: GRS test and a PCA of the five assets; compare with Lecture 3
- ▣ Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary
- ▣ Textbook for PCA: [Härdle & Simar \(2019\)](#)
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code

- ▣ **Notebook:** `chapter3_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_03

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